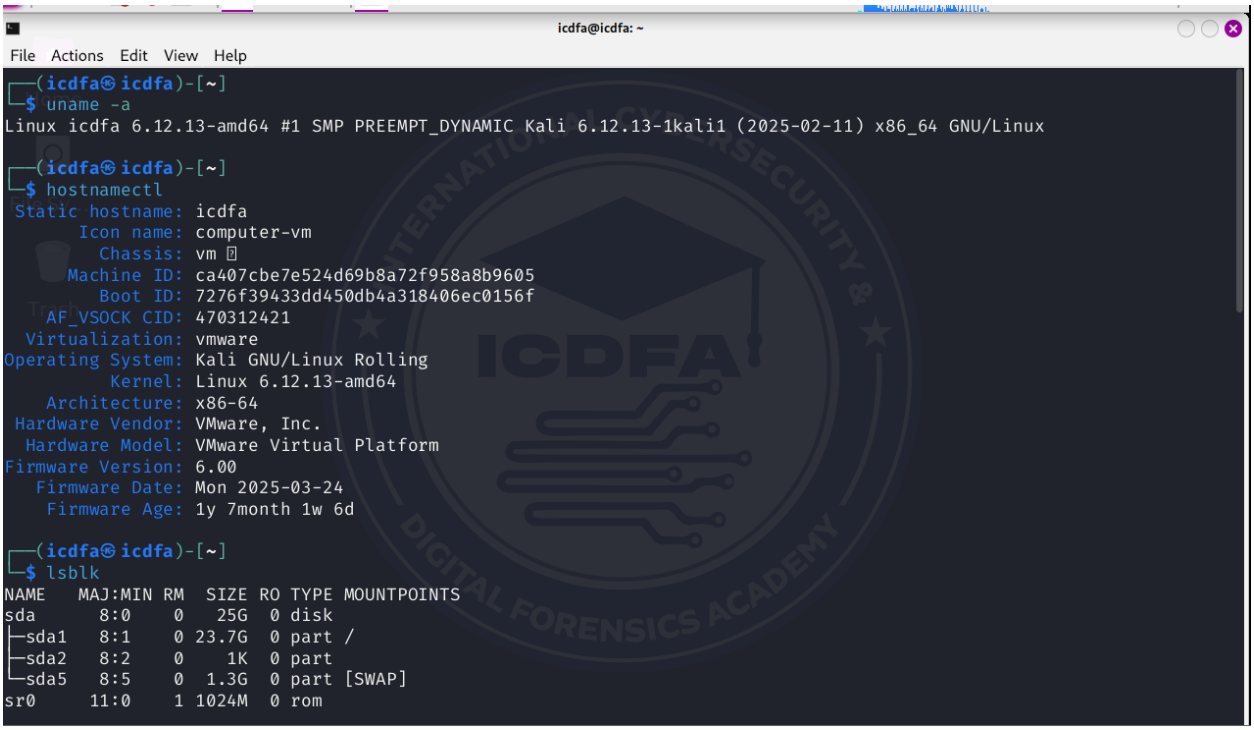
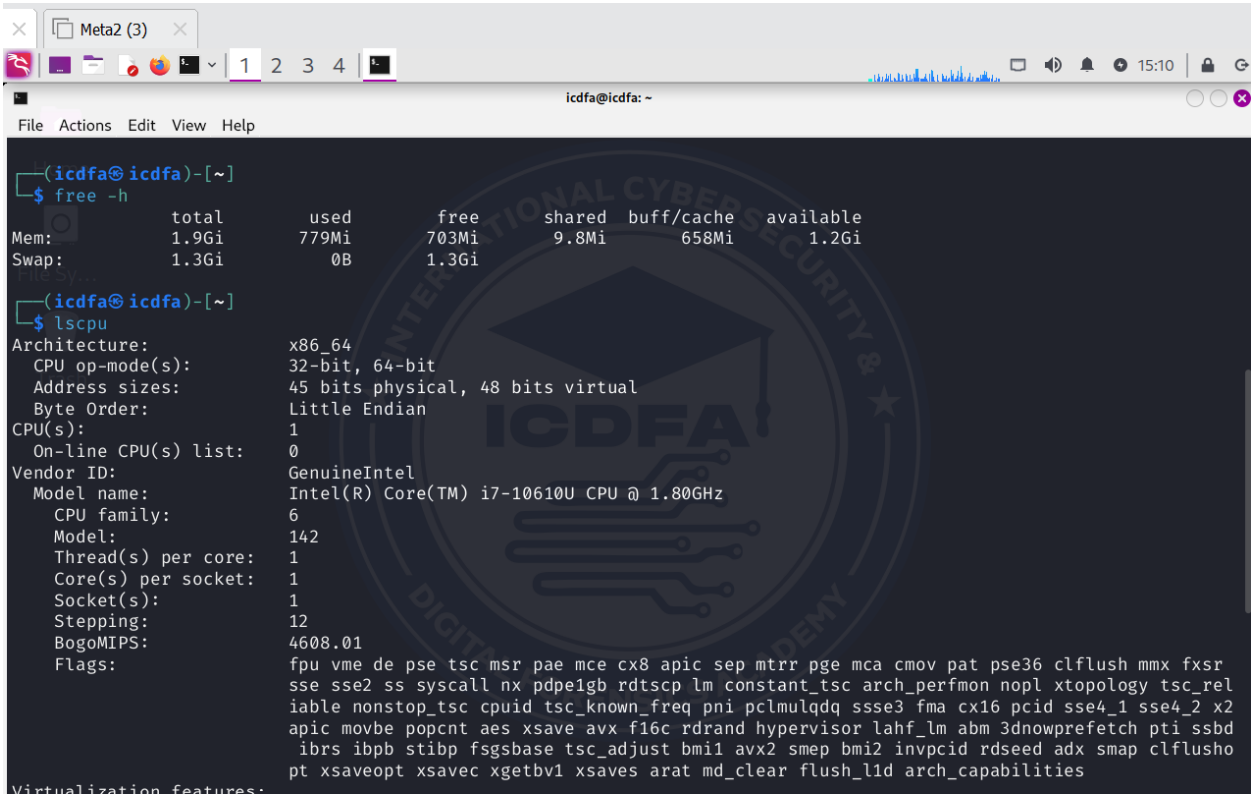
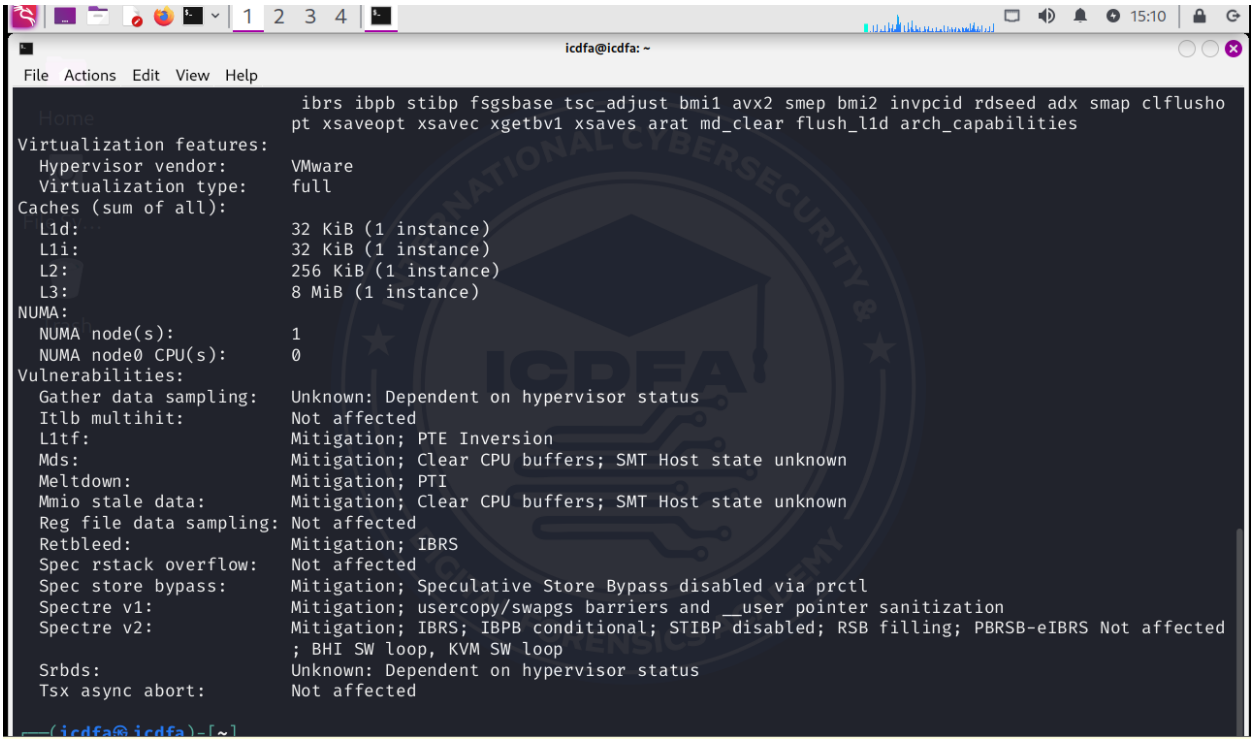
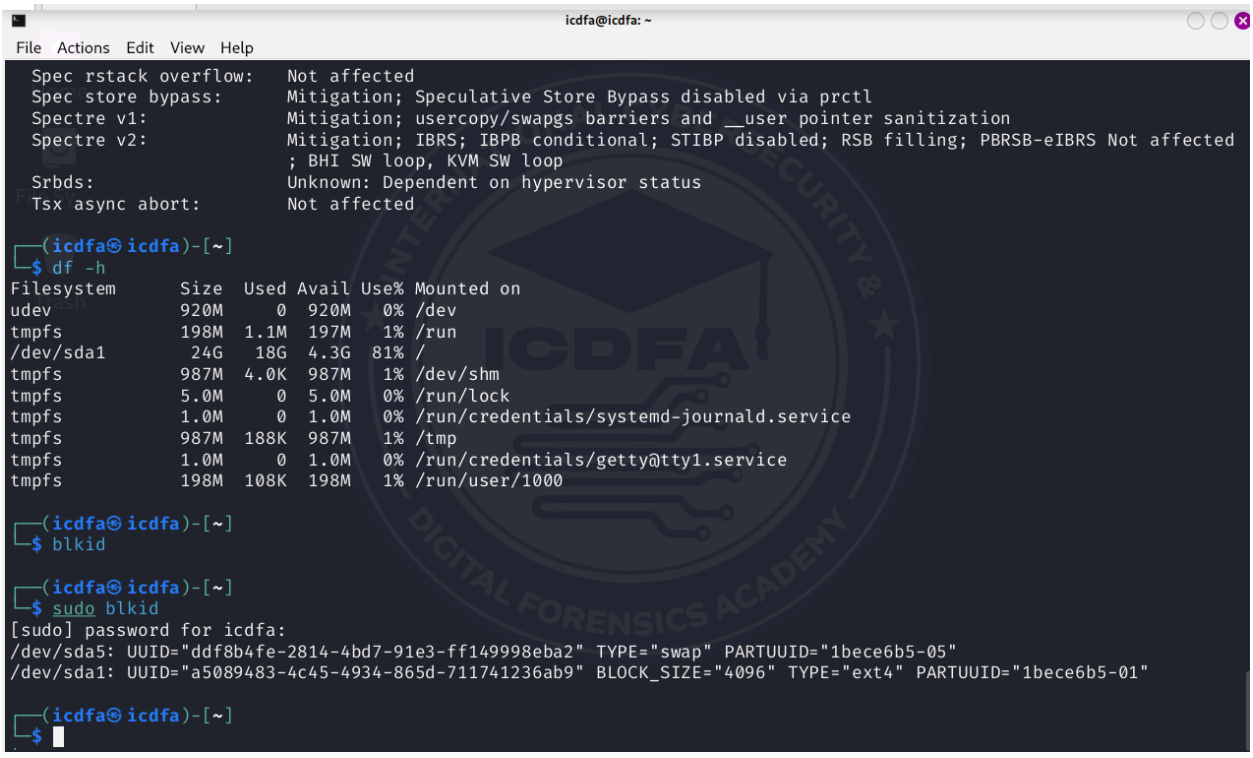
Student Name	**Maryjudith Chidinma Ogunaka**
Registration Number	**C11/26/FCDF/17151**
Programme	**ICDFA Fellowship in Cybersecurity and Digital Forensics (Cohort 11)**
Course Code	**CIP-B101**
Course Title	**Basic Computer Skills for Digital Forensics**
Lab	**Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis**
Platform	**Kali GNU/Linux Rolling virtual machine (VMware)**
Date	**05 November 2026**

Part A: System Identification and Evidence Layer Mapping






1 2 3 4
15:10
icdfa@icdfa: ~
File Actions Edit View Help
ibrs ibpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2 invpcid rdseed adx smap clflusho
pt xsaveopt xsavec xgetbv1 xsaves arat md_clear flush_l1d arch_capabilities
Virtualization features:
Hypervisor vendor: VMware
Virtualization type: full
Caches (sum of all):
L1d: 32 KiB (1 instance)
L1i: 32 KiB (1 instance)
L2: 256 KiB (1 instance)
L3: 8 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0
Vulnerabilities:
Gather data sampling: Unknown: Dependent on hypervisor status
Itlb multihit: Not affected
L1tf: Mitigation; PTE Inversion
Mds: Mitigation; Clear CPU buffers; SMT Host state unknown
Meltdown: Mitigation; PTI
Mmio stale data: Mitigation; Clear CPU buffers; SMT Host state unknown
Reg file data sampling: Not affected
Retbleed: Mitigation; IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; IBRS; IBPB conditional; STIBP disabled; RSB filling; PBRSB-eIBRS Not affected
; BHI SW loop, KVM SW loop
Srbds: Unknown: Dependent on hypervisor status
Tsx async abort: Not affected
┌──(icdfa㉿icdfa)-[~]


icdfa@icdfa: ~
File Actions Edit View Help
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; IBRS; IBPB conditional; STIBP disabled; RSB filling; PBRSB-eIBRS Not affected
; BHI SW loop, KVM SW loop
Srbds: Unknown: Dependent on hypervisor status
Tsx async abort: Not affected
┌──(icdfa㉿icdfa)-[~]
└─$ df -h
Filesystem Size Used Avail Use% Mounted on
udev 920M 0 920M 0% /dev
tmpfs 198M 1.1M 197M 1% /run
/dev/sda1 24G 18G 4.3G 81% /
tmpfs 987M 4.0K 987M 1% /dev/shm
tmpfs 5.0M 0 5.0M 0% /run/lock
tmpfs 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service
tmpfs 987M 188K 987M 1% /tmp
tmpfs 1.0M 0 1.0M 0% /run/credentials/getty@tty1.service
tmpfs 198M 108K 198M 1% /run/user/1000
┌──(icdfa㉿icdfa)-[~]
└─$ blkid
┌──(icdfa㉿icdfa)-[~]
└─$ sudo blkid
[sudo] password for icdfa:
/dev/sda5: UUID="ddf8b4fe-2814-4bd7-91e3-ff149998eba2" TYPE="swap" PARTUUID="1bece6b5-05"
/dev/sda1: UUID="a5089483-4c45-4934-865d-711741236ab9" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="1bece6b5-01"
┌──(icdfa㉿icdfa)-[~]
└─$

Part A Report Answer

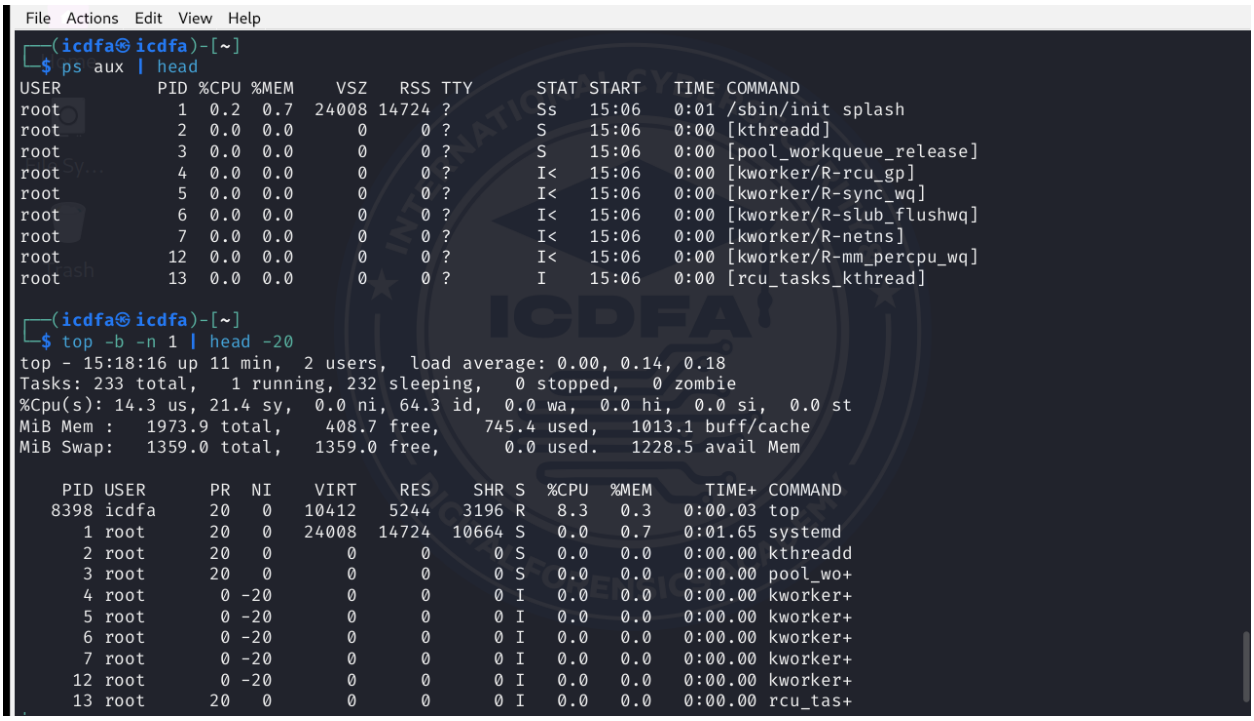
System Identification and Evidence Layer Mapping

The examination system was a VMware virtual machine running Kali GNU/Linux Rolling on an x86_64 architecture. The system utilized a 25 GB virtual disk divided into multiple partitions, including a 23.7 GB root partition and a 1.3 GB swap partition. The virtual machine was allocated approximately 2 GB of RAM. Storage analysis showed the root partition mounted at / with approximately 81% utilization.

Evidence Layer Mapping

Evidence Layer	Example Evidence	Forensic Significance
User Layer	User files, browser history, downloads	Associates' activity with a specific user
Application Layer	Browser c ache, logs, chat records	Reveals application usage and activity
Operating System Layer	Event logs, user accounts, installed software	Establishes system activity timeline
Disk Layer	Partitions, deleted files, unallocated space	Preserves persistent evidence
Memory Layer	Running processes, network sessions	Captures live system state

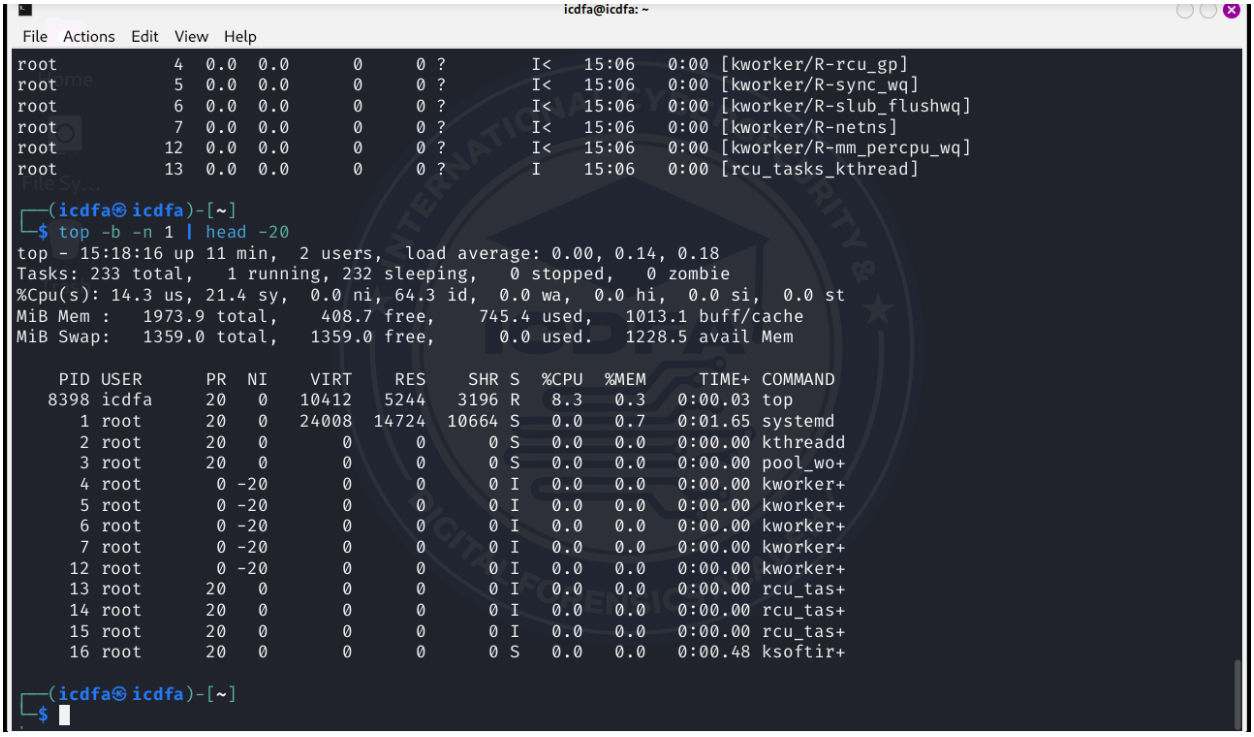
PART B: (Process Investigation)



```
File  Actions  Edit  View  Help
┌──(icdfa㉿icdfa)-[~]
└─$ ps aux | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.2  0.7  24008 14724 ?        Ss   15:06   0:01 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    15:06   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    15:06   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   15:06   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   15:06   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   15:06   0:00 [kworker/R-slub_flushwq]
root           7  0.0  0.0      0     0 ?        I<   15:06   0:00 [kworker/R-netns]
root          12  0.0  0.0      0     0 ?        I<   15:06   0:00 [kworker/R-mm_percpu_wq]
root          13  0.0  0.0      0     0 ?        I    15:06   0:00 [rcu_tasks_kthread]

┌──(icdfa㉿icdfa)-[~]
└─$ top -b -n 1 | head -20
top - 15:18:16 up 11 min,  2 users,  load average: 0.00, 0.14, 0.18
Tasks: 233 total,   1 running, 232 sleeping,   0 stopped,   0 zombie
%Cpu(s): 14.3 us, 21.4 sy,  0.0 ni, 64.3 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   1973.9 total,    408.7 free,    745.4 used,   1013.1 buff/cache
MiB Swap:   1359.0 total,   1359.0 free,      0.0 used.   1228.5 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   8398 icdfa     20   0   10412   5244   3196 R   8.3   0.3   0:00.03 top
      1 root      20   0   24008  14724  10664 S   0.0   0.7   0:01.65 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     12 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     13 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tas+
```



```
icdfa@icdfa: ~
File  Actions  Edit  View  Help
root           4  0.0  0.0      0     0 ?        I<   15:06   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   15:06   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   15:06   0:00 [kworker/R-slub_flushwq]
root           7  0.0  0.0      0     0 ?        I<   15:06   0:00 [kworker/R-netns]
root          12  0.0  0.0      0     0 ?        I<   15:06   0:00 [kworker/R-mm_percpu_wq]
root          13  0.0  0.0      0     0 ?        I    15:06   0:00 [rcu_tasks_kthread]

┌──(icdfa㉿icdfa)-[~]
└─$ top -b -n 1 | head -20
top - 15:18:16 up 11 min,  2 users,  load average: 0.00, 0.14, 0.18
Tasks: 233 total,   1 running, 232 sleeping,   0 stopped,   0 zombie
%Cpu(s): 14.3 us, 21.4 sy,  0.0 ni, 64.3 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   1973.9 total,    408.7 free,    745.4 used,   1013.1 buff/cache
MiB Swap:   1359.0 total,   1359.0 free,      0.0 used.   1228.5 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   8398 icdfa     20   0   10412   5244   3196 R   8.3   0.3   0:00.03 top
      1 root      20   0   24008  14724  10664 S   0.0   0.7   0:01.65 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     12 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     13 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tas+
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tas+
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tas+
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.48 ksoftir+

┌──(icdfa㉿icdfa)-[~]
└─$
```

Part B Report Write-up

Process Investigation

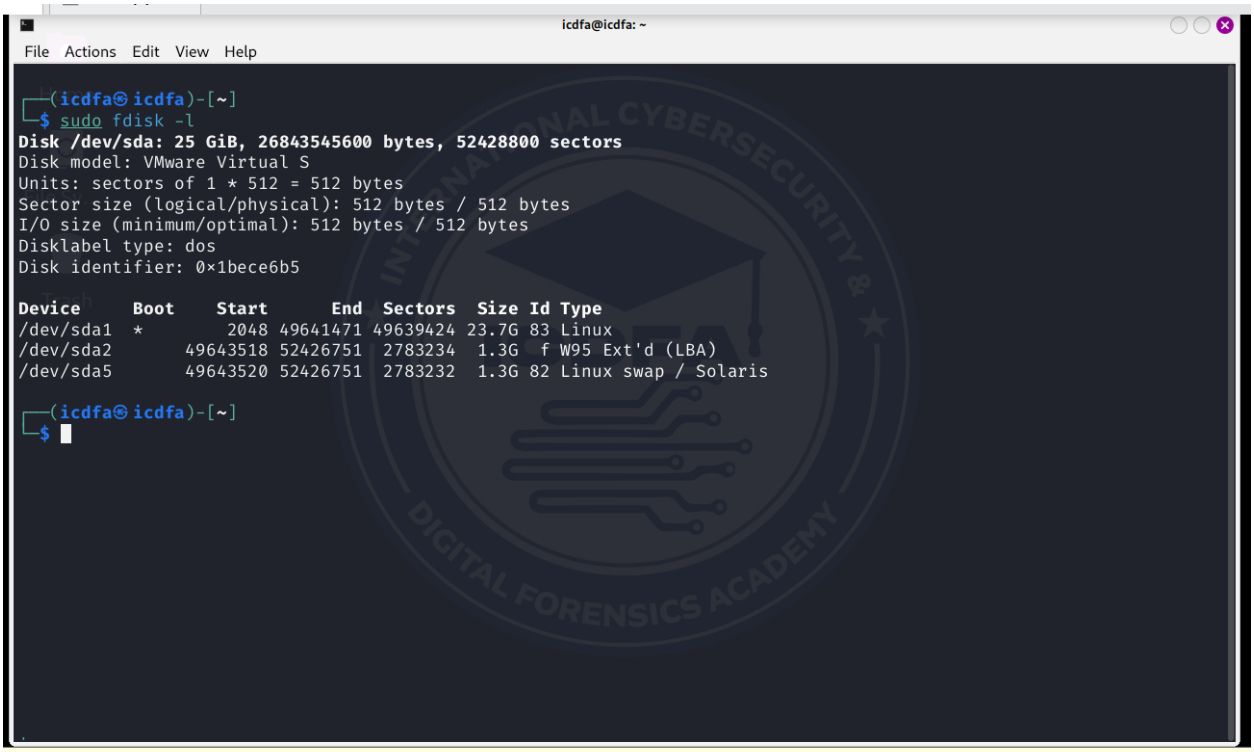
The Linux top utility was used to examine active system processes. At the time of acquisition, the system had 233 total processes, with one running process and 232 sleeping processes. No zombie processes were detected. The system had been running for approximately 11 minutes and showed minimal processor utilization.

The process list revealed standard kernel and system management processes including systemd, kthreadd, kworker, rcu_tasks, and ksoftirqd. These processes are essential for task scheduling, interrupting handling, and resource management within the Linux operating system.

Why Process Monitoring Matters in Digital Forensics

Process analysis provides insight into activities occurring on a system at the time of examination. Investigators can identify unauthorized applications, malware, suspicious scripts, persistence mechanisms, and active network connections. Memory-resident threats that may not yet be written to disk can also be discovered through process examination.

Part C (Disk Sector & Capacity Calculations)



Part C (Disk Sector & Capacity Analysis)

From the

screenshot:

- Disk: **/dev/sda**
- Capacity: **25 GiB**
- Bytes: **26,843,545,600**
- Total Sectors: **52,428,800**
- Sector Size: **512 bytes**
- Disklabel Type: **DOS (MBR)**
- Boot Partition: **/dev/sda1**

Part C Answers

1. What is the sector size?

512 bytes per sector

Sector size (logical/physical): 512 bytes / 512 bytes

2. How many sectors are on the disk?

52,428,800 sectors

3. Verify disk capacity

Total Sectors × Sector Size

= 52,428,800 × 512

= 26,843,545,600 bytes

Result: **26,843,545,600 bytes (25 GiB)**

4. Partition Table Type

Disklabel type: dos

Answer: **MBR (Master Boot Record)**

5. Which partition is bootable?

Answer:**/dev/sda1**

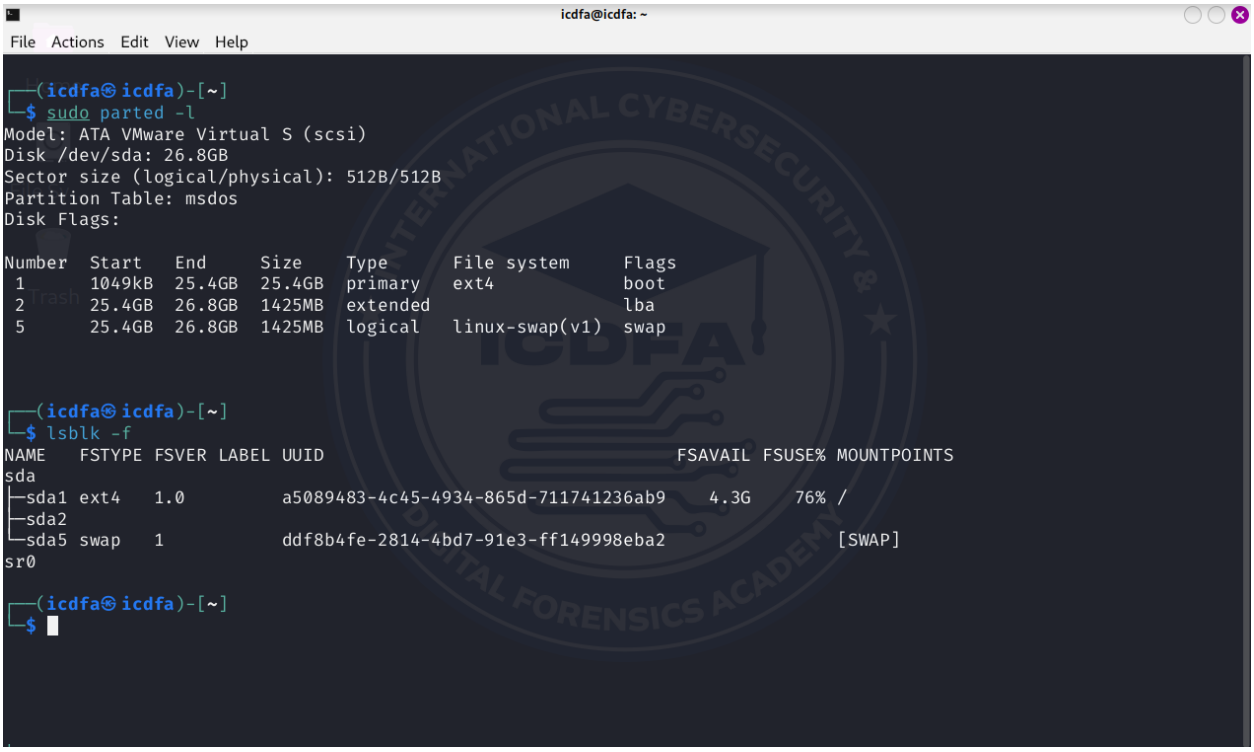
Report Writeup for Part C

Disk Sector and Capacity Analysis

The fdisk -l utility was used to inspect disk geometry and partition information. The virtual disk contained 52,428,800 sectors, each measuring 512 bytes. Multiplying the total sector count by the sector size produced a total disk capacity of 26,843,545,600 bytes (25 GiB).

The system utilized a DOS disk label format, indicating the use of the Master Boot Record (MBR) partitioning scheme. The primary bootable partition was identified as /dev/sda1, which contained the Linux operating system installation. The remaining partitions included an extended partition and a Linux swap partition.

Part D (Partition Analysis)



Excellent! ✅ **Part D (Partition Analysis)** is now complete.

Partition Structure

Partition	Type	File System	Size	Purpose
/dev/sda1	Primary	ext4	25.4 GB	Operating System Partition
/dev/sda2	Extended	N/A	1.4 GB	Container for logical partition
/dev/sda5	Logical	Linux Swap	1.4 GB	Virtual Memory (Swap Space)

Key Findings

- Partition Table Type: **msdos (MBR)**

- Main filesystem: **ext4**
- Swap filesystem: **linux-swap(v1)**
- Boot flag present on **/dev/sda1**
- Root filesystem mounted at **/**

Part D Report Write-up

Partition Analysis

The parted and lsblk utilities were used to examine the disk partition layout and filesystem configuration. The system utilized an MBR (msdos) partitioning scheme consisting of one primary partition, one extended partition, and one logical partition.

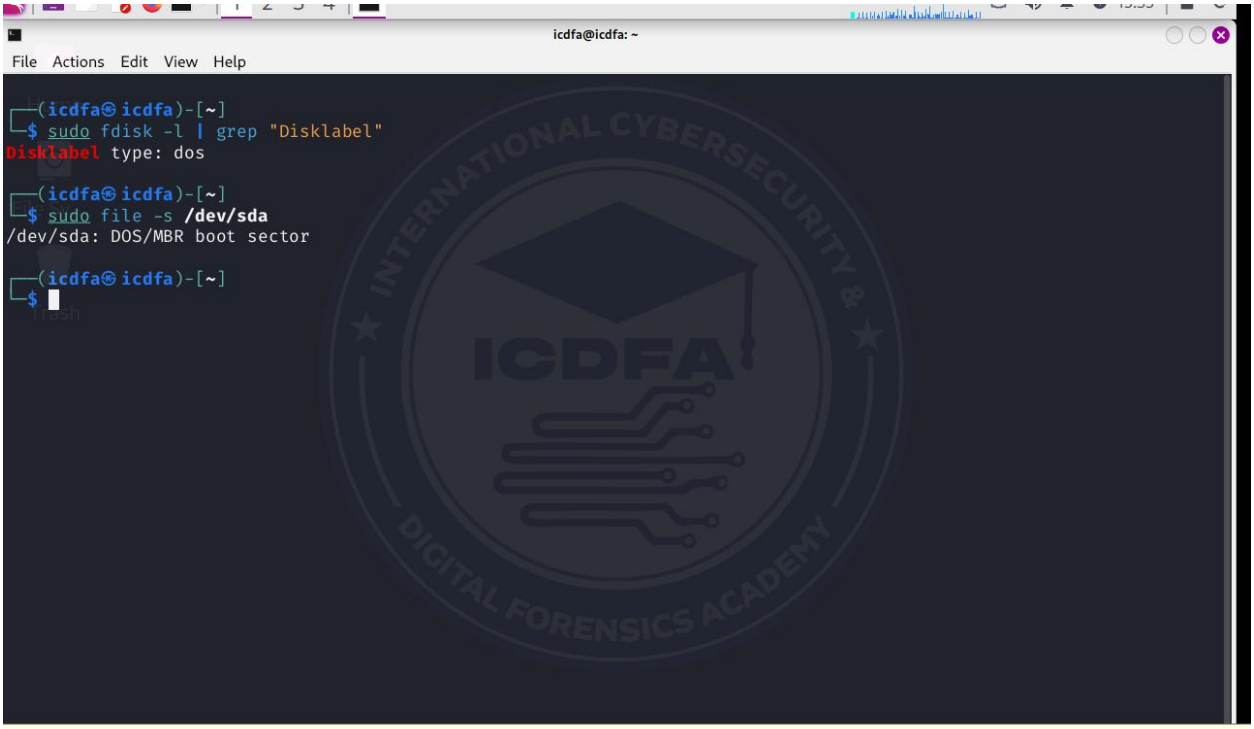
The primary partition (/dev/sda1) was formatted using the ext4 filesystem and served as the operating system partition. The extended partition (/dev/sda2) contained a logical swap partition (/dev/sda5) used for virtual memory management.

The boot flag was assigned to the primary partition, indicating that it contains the files required to start the operating system.

Forensic Importance

Partition analysis helps investigators identify storage locations, understand disk organization, locate operating systems, and determine where potential evidence may reside. Knowledge of filesystem types assists in selecting appropriate forensic acquisition and analysis techniques.

Part E (Boot Process, MBR and GPT Concepts)



Part E Report Write-up

Boot Process and Partition Table Analysis

The Linux utilities fdisk and file were used to examine the system's boot structure. Analysis identified the disk label type as DOS, which corresponds to the Master Boot Record (MBR) partitioning scheme.

Further verification using the file utility identified /dev/sda as a DOS/MBR boot sector, confirming the presence of boot code within the first sector of the disk.

What is MBR?

The Master Boot Record occupies the first sector of a storage device (Sector 0) and contains:

1. Bootstrap code used during system startup.
2. The partition table.
3. The boot signature.

When a computer starts, the BIOS loads the MBR into memory and executes the boot code, which then locates the active operating system partition.

Forensic Significance

The MBR is a critical forensic artifact because malware can modify boot records to establish persistence. Examiners often validate the integrity of the MBR during forensic investigations to identify unauthorized modifications or bootkits.



Part F (File System Analysis) is complete.

Evidence Collected

From the screenshot:

- /dev/sda1 uses the **ext4** filesystem.
- /dev/sda1 is mounted on the root directory (/).
- The filesystem is mounted with read-write permissions (rw).
- Disk utilization is approximately **81%.**
- Temporary filesystems (tmpfs) are used for memory-based storage.

- The root filesystem is the primary location where operating system and user data are stored.

Part F Report Write-up

File System Analysis

The df -T and mount commands were used to identify the filesystem types and mounted storage devices on the examination system.

Analysis showed that the primary partition (/dev/sda1) was formatted using the **ext4 filesystem**, which is the default filesystem used by many Linux distributions. The partition was mounted as the root file system (/) and configured with read-write access.

The ext4 filesystem provides journaling capabilities that improve reliability and help maintain filesystem integrity following unexpected shutdowns or system failures.

Forensic Significance

Understanding the filesystem is important because it determines how files, timestamps, permissions, and metadata are stored. In forensic examinations, investigators analyze filesystem structures to recover deleted files, identify user activity, and reconstruct timelines of events. The ext4 filesystem stores valuable metadata that can assist in determining file creation, modification, and access patterns.